

Answers to Exercises

$$1. \quad 2 + 2i = 8^{1/2}e^{i(\pi/4)} \implies (2 + 2i)^6 = \left(8^{1/2}e^{i(\pi/4)}\right)^6 = 8^{6/2}e^{i(6\pi/4)} = 8^3e^{i(3\pi/2)}$$

$$2. \quad 2 + 2i = 8^{1/2}e^{i(\pi/4)} \implies (2 + 2i)^{1/3} = \left(8^{1/2}e^{i(\pi/4)}\right)^{1/3} = 8^{1/6}e^{i(\pi/12+2\pi k/3)} \quad k = 0, 1, 2$$

$$k = 0 : \quad 8^{1/6}e^{i(\pi/12)}$$

$$k = 1 : \quad 8^{1/6}e^{i(9\pi/12)}$$

$$k = 2 : \quad 8^{1/6}e^{i(17\pi/12)}$$

$$3. \quad 4(1 - \sqrt{3}i) = 8e^{i(5\pi/3)} \text{ or } 8e^{i(-\pi/3)} \implies \left(4(1 - \sqrt{3}i)\right)^7 = \left(8e^{i(-\pi/3)}\right)^7 = 8^7e^{i(-7\pi/3)}$$

$$4. \quad 4(1 - \sqrt{3}i) = 8e^{i(5\pi/3)} \text{ or } 8e^{i(-\pi/3)} \implies \left(4(1 - \sqrt{3}i)\right)^{1/4} = \left(8e^{i(-\pi/3)}\right)^{1/4} = 8^{1/4}e^{i(-\pi/12+2\pi k/4)}$$

$$k = 0, 1, 2, 3$$

$$k = 0 : \quad 8^{1/4}e^{i(-\pi/12)}$$

$$k = 1 : \quad 8^{1/4}e^{i(5\pi/12)}$$

$$k = 2 : \quad 8^{1/4}e^{i(11\pi/12)}$$

$$k = 3 : \quad 8^{1/4}e^{i(17\pi/12)}$$

$$5. \quad -2 + 2i = 8^{1/2}e^{i(3\pi/4)} \text{ and } 3i = 3e^{i(\pi/2)}$$

$$\implies \left(\frac{-2 + 2i}{3i}\right)^{1/2} = \left(\frac{8^{1/2}e^{i(3\pi/4)}}{3e^{i(\pi/2)}}\right)^{1/2} = \left(\frac{8^{1/2}}{3}e^{i(\pi/4)}\right)^{1/2} = \frac{8^{1/4}}{3^{1/2}}e^{i(\pi/8+2\pi k/2)} \quad k = 0, 1$$

$$k = 0 : \quad \frac{8^{1/4}}{3^{1/2}}e^{i(\pi/8)}$$

$$k = 1 : \quad \frac{8^{1/4}}{3^{1/2}}e^{i(9\pi/8)}$$

$$6. \quad z^{1/6} - 2 - 2i = 0 \implies z = (2 + 2i)^6 = 8^3e^{i(3\pi/2)} \quad (\text{from 1.})$$

$$7. \quad z^3 + 5i = 2 + 7i \implies z = (2 + 2i)^{1/3} = 8^{1/6}e^{i(\pi/12+2\pi k/3)} \quad k = 0, 1, 2 \quad (\text{from 2.})$$

$$8. \quad 3iz^2 + 2 = 2i \implies z = \left(\frac{-2 + 2i}{3i}\right)^{1/2} = \frac{8^{1/4}}{3^{1/2}}e^{i(\pi/8+2\pi k/2)} \quad k = 0, 1 \quad (\text{from 5.})$$